

Memory

Introduction to Brain and Cognition

Spring 2026

Memory

- Memory - The ability to store and retrieve information over time
 - No, they are **not reproduced** like a tape playing
 - Memories are **actively** constructed and **reconstructed**
- Key functions
 - **Acquisition / Encoding** - The process by which we transform what we perceive, think or feel into an enduring memory
 - **Storage** - The process of maintaining information in memory over time
 - **Retrieval** - The process of bringing information to mind that has been previously **encoded** and **stored**

Clive Wearing - The Man with 7 second memory

- Talented musician
- Viral encephalitis at age 47
- Suffers from Amnesia
- Moment-to-moment consciousness

<https://www.youtube.com/watch?v=Vwigmktix2Y>



How does he still recognize his wife? - Emotions?
How does he still play the piano so well? - Procedural/skill memory?
Is he happy?

Brain Regions Involved in Memory Formation

Hippocampus plays a part in memory consolidation: the process of transferring new learning into long-term memory (**indexing** - Library analogy)

Engram - group of neurons that serve as “**physical representation of memory**”

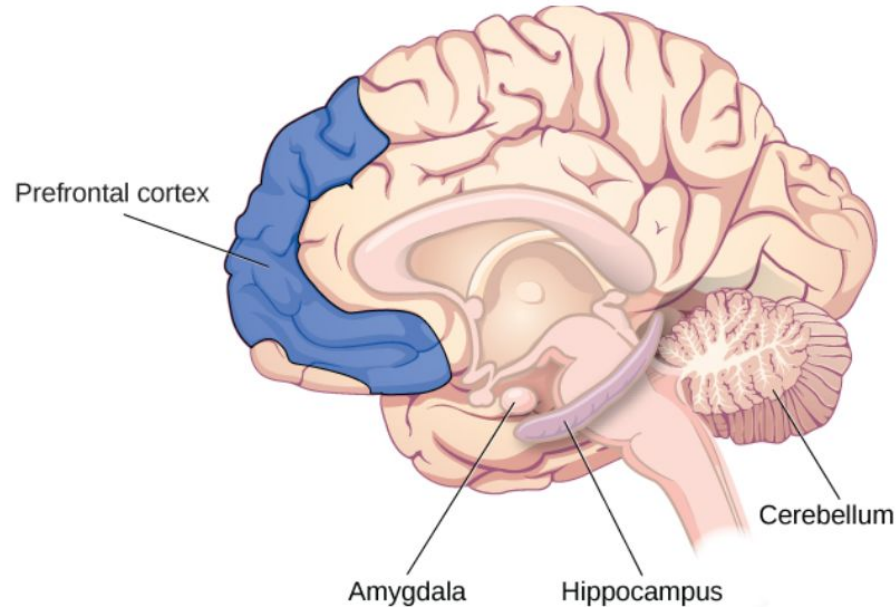


Figure 1. The amygdala is involved in fear and fear memories. The hippocampus is associated with declarative and episodic memory as well as recognition memory. The cerebellum plays a role in processing procedural memories, such as how to play the piano. The prefrontal cortex appears to be involved in remembering semantic tasks.

How is information **acquired**?

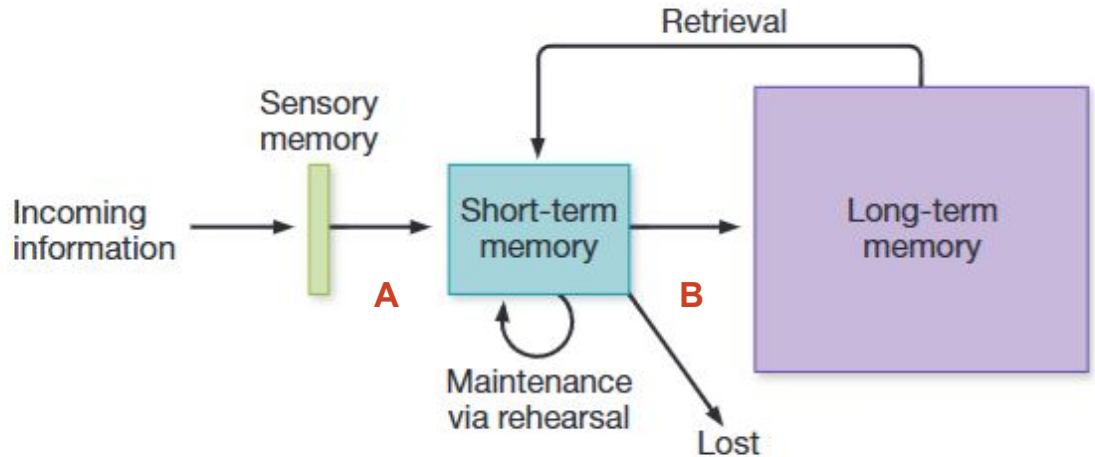
Atkinson and Shiffrin
(1968) proposed the
Modal Model

**A - Selective Attention/
Interpretation**

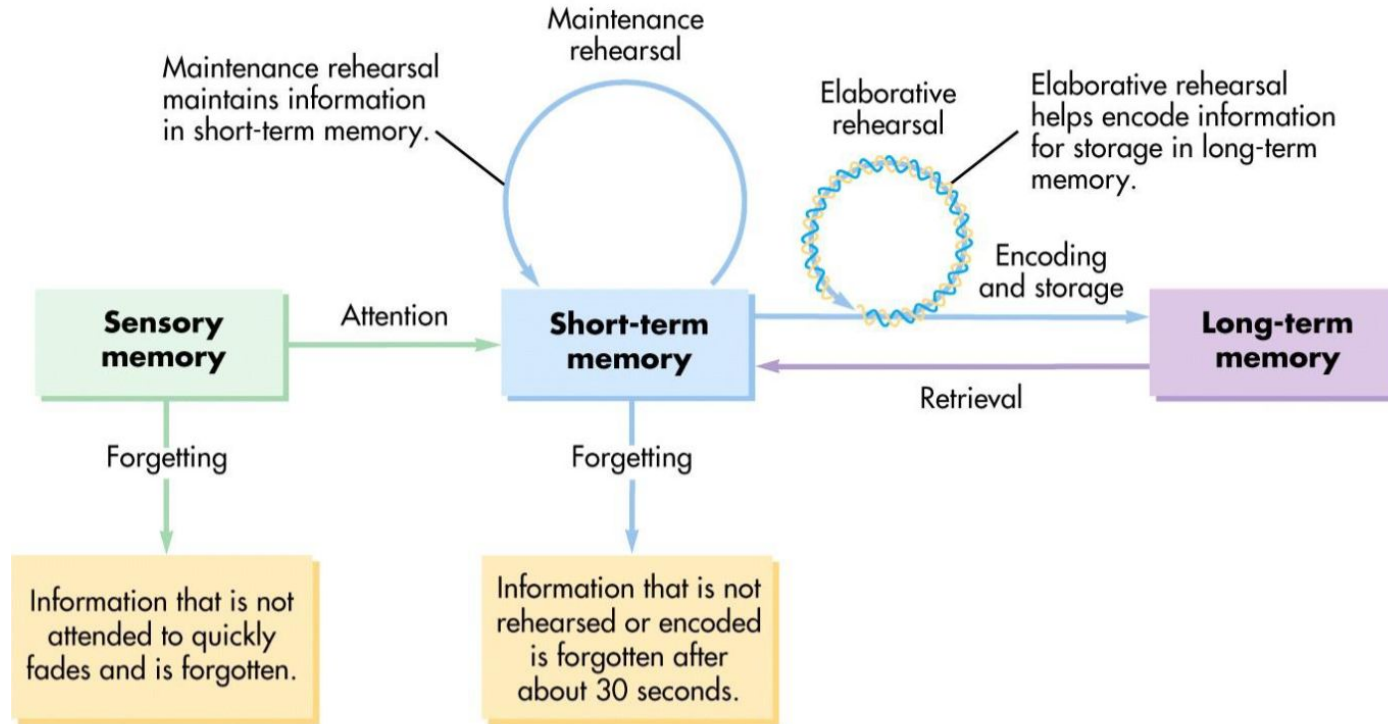
**B - Elaborative
Rehearsal**

**STM - now preferred
Working Memory**

FIGURE 6.1 AN INFORMATION-PROCESSING VIEW
OF MEMORY



Three System Memory



Storage - Maintaining information over time

Three major kinds of memory storage:

- **Sensory Memory** - Storage that holds sensory information for few seconds or less
 - **Iconic Memory** (Visual Sensory Memory) - Holds an image for about a second
 - **Echoic Memory** (Auditory Sensory Memory) - Holds sound info from a few to several seconds
- Short-Term Memory (STM)
- Long-Term Memory (LTM)

I	H	B
D	A	G
E	J	C

Test Your Iconic Memory (Sperling Task, 1960)

- Without tone - Barely 3 or 4 letters retrieved (iconic memory faded before reporting)
- With tone (High, Mid, Low) - Cue signaling helped retrieve almost all letters of a row

<https://www.youtube.com/watch?v=GkZNH49GcA&t=34s>

Short-Term Memory

Short-Term Memory - Storage that holds non-sensory information for more than a few seconds but less than a minute; can hold about 7 items

- **Rehearsal** - The process of keeping information in STM by mentally repeating it
- **Span** - Immediate memory span i.e. max # of items you can recall perfectly after seeing once
 - **Digit Span - Magic Number: 7 ± 2** meaningful items (Miller, 1956)
 - **Enlarge** - Chunking
- **Chunking** - Combining small pieces of information into larger clusters that are more easily held in STM
 - Increase the amount of information held in STM
 - GROUPINGBYMEANINGFULUNIT (GROUPING BY MEANINGFUL UNIT)
- **Duration** - Less than 20s without rehearsal (Peterson & Peterson, 1959)

Long Term Memory

- **LTM** - Storage that holds information for hours, days, weeks, years, decades; no known capacity
- Permastore (Hold information indefinitely)
- Expectations affect what is **recalled** - active reconstruction
- Jill Price - remembers every day of her life in great details! (Highly Superior Autobiographical Memory - HSAM)
- **Eyewitness testimony** - prone to distortions

Rehearsal Types

1. Maintenance Rehearsal

- a. Only focus on to-be-remembered items
- b. Without understanding their meaning or how they relate to each other
- c. Rote - mechanical process of recycling items in **WM** by repeating them over and over
- d. Inefficient (sheer exposure not sufficient to engage LTM)

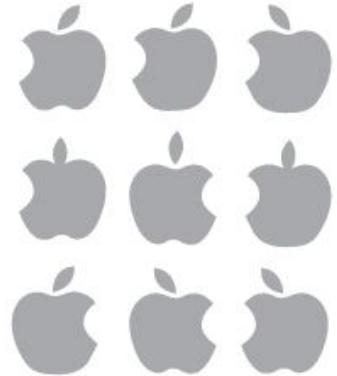
2. Relational or Elaborative Rehearsal

- a. Understanding what the items mean or how they relate to each other
- b. Superior to former rehearsal type
- c. Engages **LTM**

FIGURE 6.10 MEMORY FOR AN OFTEN-VIEWED LOGO

Most people have seen the Apple logo countless times, but they've had no reason to pay attention to its features. As a result, they have poor memories for the features. Test yourself. Can you find the correct version among the options displayed here?

(THE ANSWER IS AT THE END OF THE CHAPTER.)



Three System Memory (Summary)

	Sensory Memory	Short-term Memory	Long-term Memory
Function	Temporary storage of sensory info	Storage of info currently being used	Storage of info indefinitely
Span	High	Limited (*Chunking)	Unlimited
Duration	Less than 1 sec or a few sec	Less than 20 sec	For a lifetime

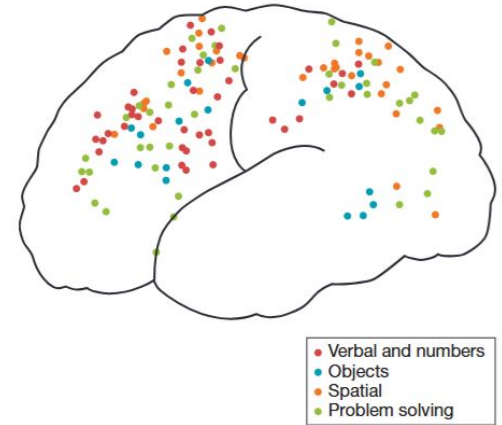
Working Memory

- Active maintenance of information in STM (**Modal Model**)
- More dynamic best understood as *status* (i.e. *currently activated*) rather than a *place* (Christophel et al. 2017)
- **Digit Span**
- **Operation Span** - Working memory capacity (WMC)
 - **Reading Span**

Due to his gross inadequacies, his position as director was terminated abruptly.

It is possible, of course, that life did not arise on Earth at all.
- People with **larger span** (greater WMC) do better in tasks requiring coordination of different pieces of info - link to **intelligence**

FIGURE 6.6 IS WORKING MEMORY A “PLACE”?



Modern theorists argue that working memory is not a place at all, but is instead the name we give for a certain set of mental activities. Consistent with this modern view, there's no specific location within the brain that serves as working memory. Instead, working memory is associated with a wide range of brain sites, as shown here. (AFTER CABEZA & NYBERG, 2000)

Baddeley & Hitch's WM Model

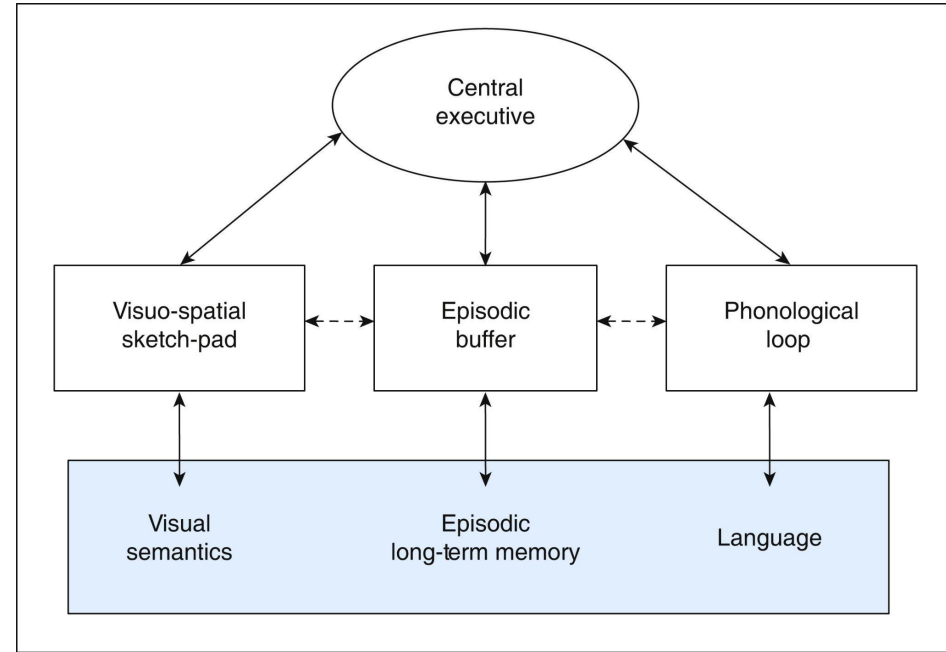
Dynamic Model: Unlike passive short-term memory, working memory actively processes and organizes information

Multi-Component System: Working memory is not a single store but consists of multiple components

- **Central Executive:** Directs attention, manages cognitive tasks, and coordinates information processing, goal planning and execution
- **Phonological Loop:** Stores and rehearses verbal information using subvocalization ("inner voice" and "inner ear") - Articulatory rehearsal loop, engages brain regions responsible for speech and hearing
- **Visuospatial Sketchpad:** Holds and manipulates visual and spatial information, like mental imagery
- **Episodic Buffer:** Integrates information across components, helps keep track of chronological order and links to long-term memory

Amnesia (Evidence for episodic buffer): Short term recall intact, but link to LTM disrupted

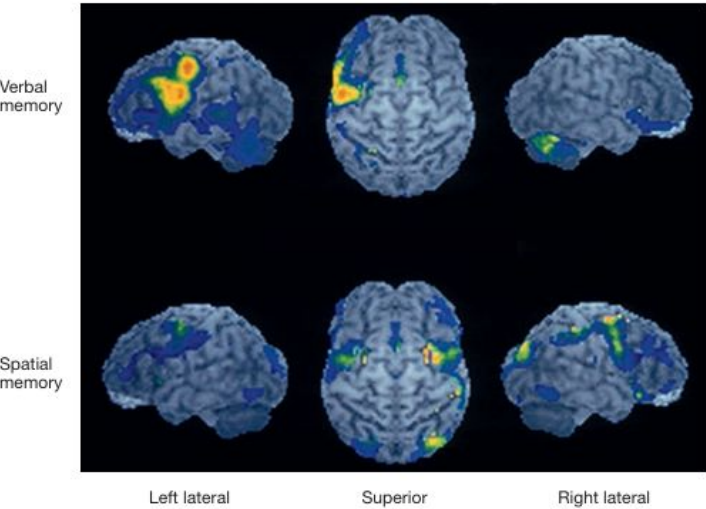
Effect of Concurrent Articulation (Evidence for phonological loop): Repeating 'tah tah tah' cuts memory span, prevents subvocal "inner voice" rehearsal



Current Biology

<https://doi.org/10.1016/j.cub.2009.12.014>

FIGURE 6.8 BRAIN ACTIVITY AND WORKING-MEMORY REHEARSAL



Color is used here as an indication of increased brain activity (measured in this case by positron emission tomography). When research participants are doing a verbal memory task (and using the articulatory loop), activation increases in areas ordinarily used for language production and perception. A very different pattern is observed when participants are doing a task requiring memory for spatial position. Notice, then, that the “inner voice” and “inner ear” aren’t casual metaphors; instead, they involve mechanisms that are ordinarily used for overt speech and actual hearing.

(AFTER JONIDES, LACEY, & NEE, 2005; ALSO SEE JONIDES ET AL., 2008)

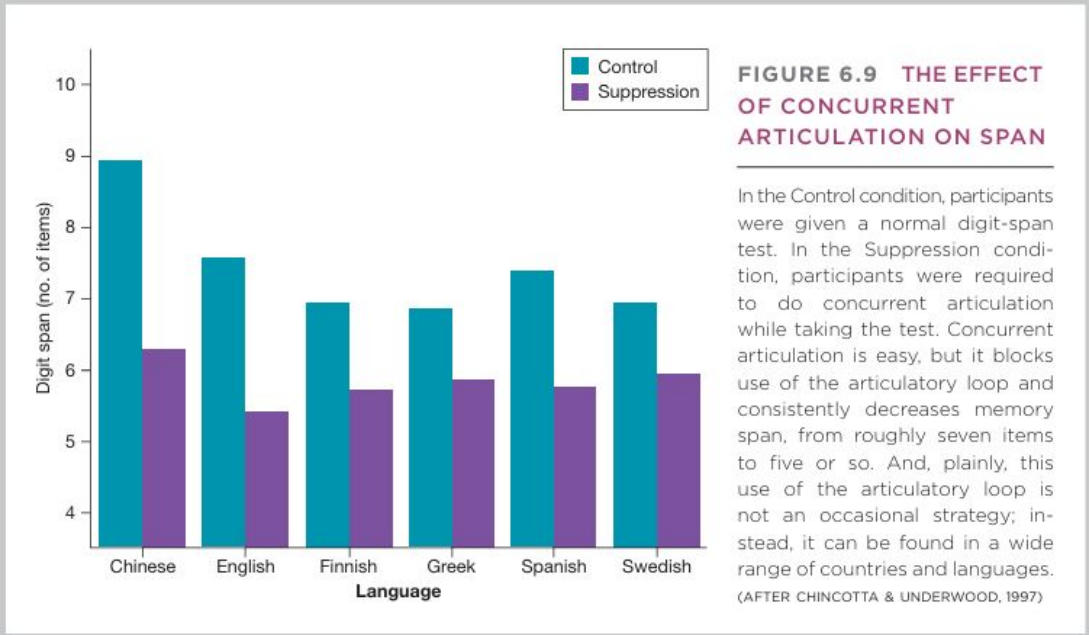


FIGURE 6.9 THE EFFECT OF CONCURRENT ARTICULATION ON SPAN

In the Control condition, participants were given a normal digit-span test. In the Suppression condition, participants were required to do concurrent articulation while taking the test. Concurrent articulation is easy, but it blocks use of the articulatory loop and consistently decreases memory span, from roughly seven items to five or so. And, plainly, this use of the articulatory loop is not an occasional strategy; instead, it can be found in a wide range of countries and languages. (AFTER CHINCOTTA & UNDERWOOD, 1997)

Primacy and Recency Effects

Primacy Effect

- More repetitions (via rehearsal strategies) increase likelihood of long-term memory (LTM) storage.
- Less competition for cognitive resources in the beginning.

Recency Effect

- Last few items remain in working memory at recall.
- No new information to push them out.
- Disappears if a delay (e.g., counting backward) prevents immediate recall.

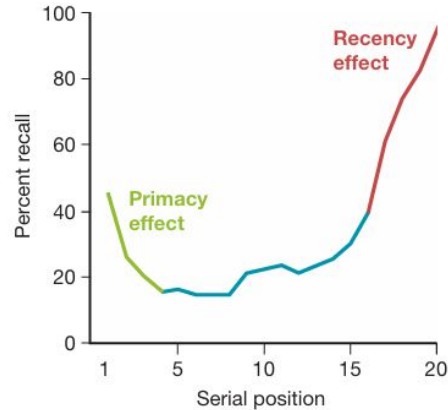


FIGURE 6.2 PRIMACY AND RECENCY EFFECTS IN FREE RECALL

Research participants in this study heard a list of 20 common words presented at a rate of one word per second. Immediately after hearing the list, participants were asked to write down as many of the words on the list as they could recall. The results show that position in the series strongly affected recall—participants had better recall for words at the beginning of the list (the primacy effect) and for words at the end of the list (the recency effect), compared to words in the middle of the list.

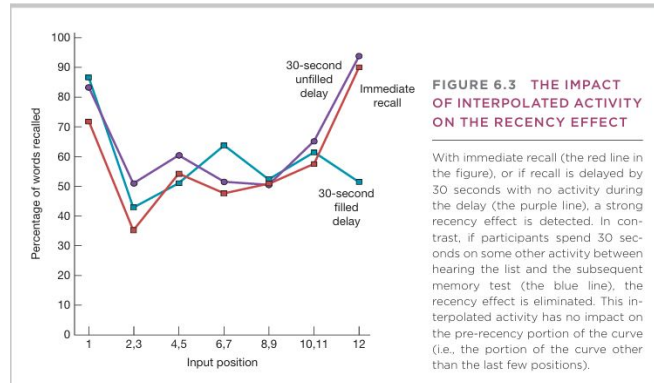


FIGURE 6.3 THE IMPACT OF INTERPOLATED ACTIVITY ON THE RECENCY EFFECT

With immediate recall (the red line in the figure), or if recall is delayed by 30 seconds with no activity during the delay (the purple line), a strong recency effect is detected. In contrast, if participants spend 30 seconds on some other activity between hearing the list and the subsequent memory test (the blue line), the recency effect is eliminated. This interpolated activity has no impact on the pre-recency portion of the curve (i.e., the portion of the curve other than the last few positions).

Intentional Learning Vs Incidental Learning

Intentional Learning

- Deliberate learning, with an expectation that memory will be tested later. Conscious effort is put into memorizing and learning something.
- Memory can be just as effective without the intention to learn, as long as the material is processed in an effective manner.

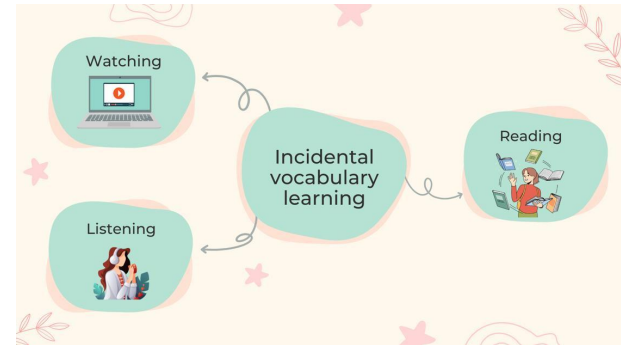
Incidental Learning

- Occurs without any conscious effort. Unintentional.

Levels of Processing Framework (Craik and Lockhart, 1972) - Memory retention depends on the level at which information is processed during encoding.

- Depth of Processing - Shallow, Deep

Second Language Vocabulary Acquisition Incidental or Intentional Learning?



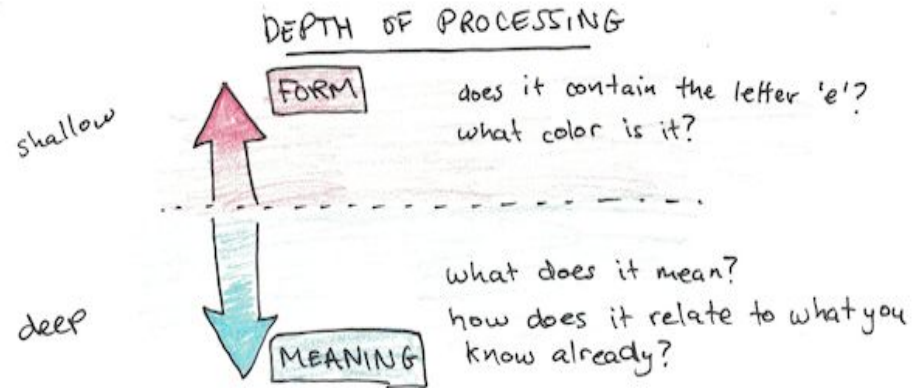
Depth of Processing

Shallow processing

Involves processing information **superficially**, focusing on its surface features such as its physical appearance, structure, or sound. It typically involves encoding information based on its **perceptual features** rather than its meaning or significance.

Deep processing

Involves encoding information based on its **meaning, relevance, and significance**. It requires **more cognitive effort and engagement**, as it involves analyzing and understanding the content of the information.

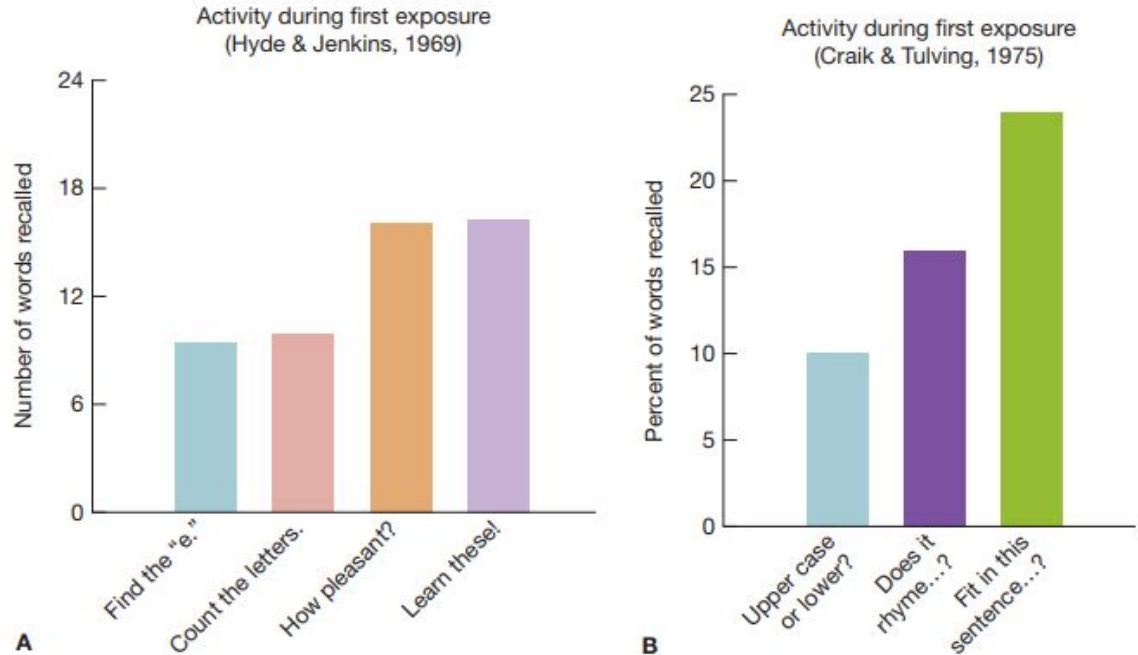


Depth of Processing

Shallow processing experiment tasks like find the 'e' or counting letters or identifying capital letters: Lower memory retention

Deep processing task like does it rhyme, learn these involve more processing: Better memory retention

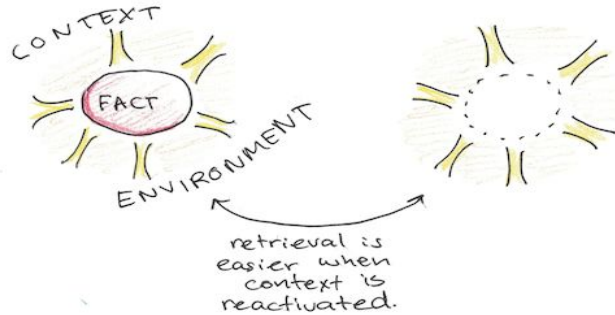
FIGURE 6.12 THE IMPACT OF DEEPER PROCESSING



The Role of Context

Encoding Specificity: Memory is specific, encompassing not just the stimulus itself but also its context and associations.

Context-Dependent Learning: Memory is influenced by the context in which information is encoded and retrieved.



State-Dependent Learning: Memory retrieval is influenced by an individual's internal state or physiological condition during encoding and retrieval.



FIGURE 7.2 THE DESIGN OF A CONTEXT-DEPENDENT LEARNING EXPERIMENT

Half of the participants (deep-sea divers) learned the test material while underwater; half learned while on land. Then, within each group, half were tested while underwater; half were tested on land. We expect a retrieval advantage if the learning and test circumstances match. Therefore, we expect better performance in the top left and bottom right cells.

		Test while	
		On land	Underwater
Learn while	On land	Learning and test circumstances match	<i>CHANGE</i> of circumstances between learning and test
	Underwater	<i>CHANGE</i> of circumstances between learning and test	Learning and test circumstances match

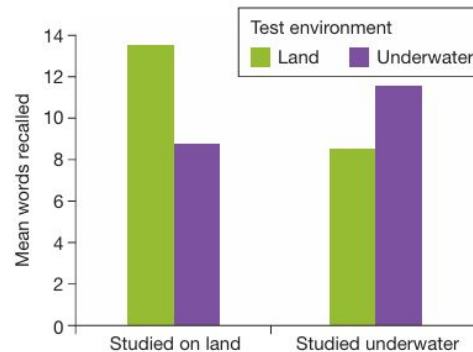


FIGURE 7.3 CONTEXT-DEPENDENT LEARNING

Scuba divers learned materials either while on land or while underwater. Then, they were tested while on land or underwater. Performance was best if the divers' circumstances at the time of test were matched to those in place during learning. (AFTER GODDEN & BADDELEY, 1975)

Elaborate Processing vs Deep & Elaborate Encoding

Craik & Tulving, 1975 study:

Shown a word like chicken

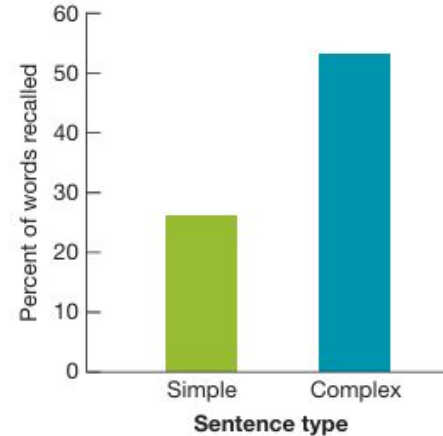
Now, decide if this word fits into this sentence:

She cooked the -----

Consider a more complex sentence: The great bird swooped down and carried away the -----

Multiple connections! Multiple retrieval paths.

FIGURE 6.13 DEEP AND ELABORATE ENCODING



Deep processing (paying attention to meaning) promotes memory, but it isn't the only factor that has this benefit. More elaborate processing (e.g., by thinking about the word in the context of a complex sentence, rich with relationships) also has a powerful effect on memory. (AFTER CRAIK & TULVING, 1975)

Organizing Memory & advantages/limitations

1. Power of mnemonics
2. Understanding
 - a. Consolidation
3. Rote learning
4. Spaced learning
5. Massed learning
 - a. No reconsolidation (you need sleep)

FIGURE 6.17 COMPREHENSION ALSO AIDS MEMORY FOR PICTURES



People who perceive this picture as a pattern of meaningless blotches are unlikely to remember the picture. People who perceive the "hidden" form do remember the picture. (AFTER WISEMAN & NEISSER, 1974)